PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field 💫



https://github.com/SuleynanAuir/PEANUT--Prompt-Enhanced-Ablation-with-Optical-Flow-Based-Neural-Unit-from-Image2Video-Field

PART 01

Introduction

Presentor： Huang Siqi (2330034020)

PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field

Two Main Problems of Existing Technologies

1) Temporal Consistency Issues

- Texture Swimming: Background details appear unstable
- Edge Jitter/Fragmentation



Root Causes:
Local optimal (Error Propagation Across Frames)

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2) Artifacts in Complex Motion





Root Causes:
Reference the wrong frame during adjacent frame alignment

Goal 1: Free your hands and time

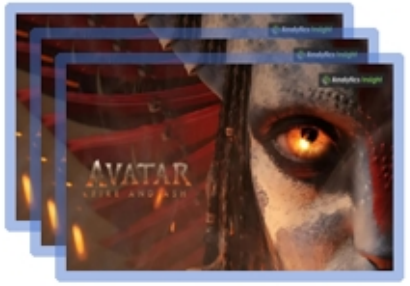


Just Tap in the Words U want to remove!

Image



Video Frame

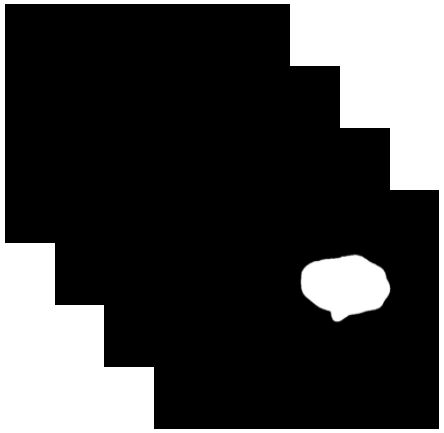


Prompt Block

P_1: *the* ***red eyes***

P_2: *ones with* ***anger***

P-MASk Module



Prompt-Guided Mask Generation (P- MASk)

Goal 2: Lightweighted Deployment

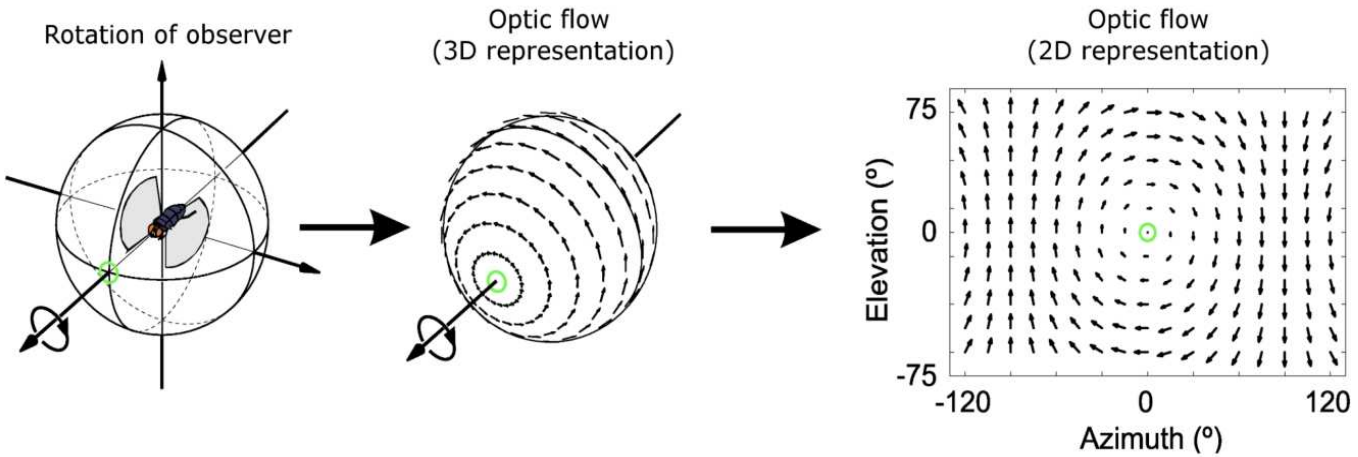
Goal 3: One-Stop Service



Collection by wanqiaox

Lightweight models

Traditional Fluid-based Methods
+
Modern Deep Learning



Optical Flow

- Feature Propagation
- Deformable Convolution
- Deformable Convolution
- Content Hallucination Self-Attention Module

Network

What u need to do: **Just type in Removal Requirements & Load Video**

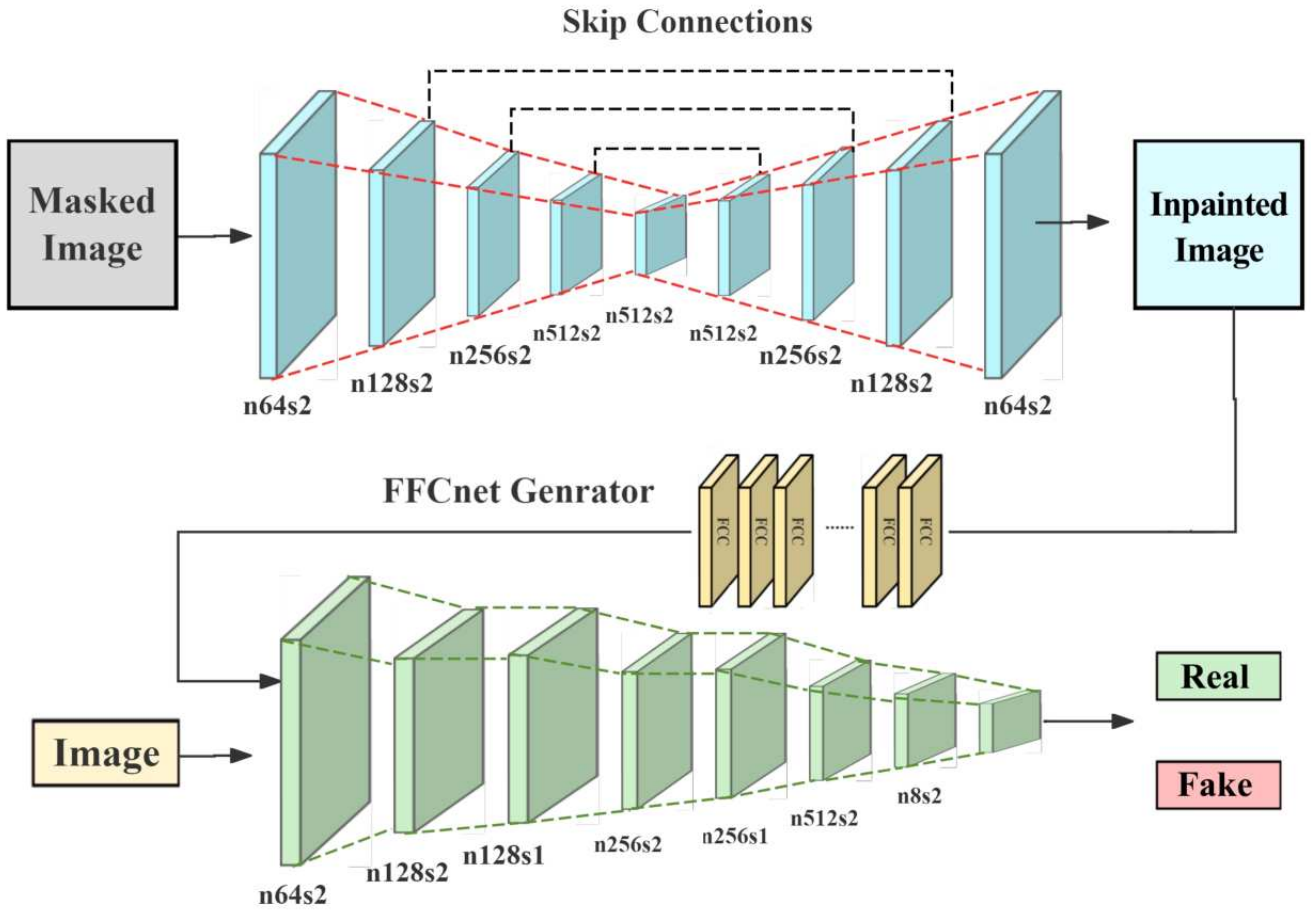
PART 02

Big-LaMa in Video World

Presentor： Liang Yirui (2330034030)
Chen Qianqi (233034001)

PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field

2.1 Big-LaMa Technology & Benefits



Structure of Big-LaMa:
1. Encoder–Decoder
2. Skip Connections
3. Multi-scale features

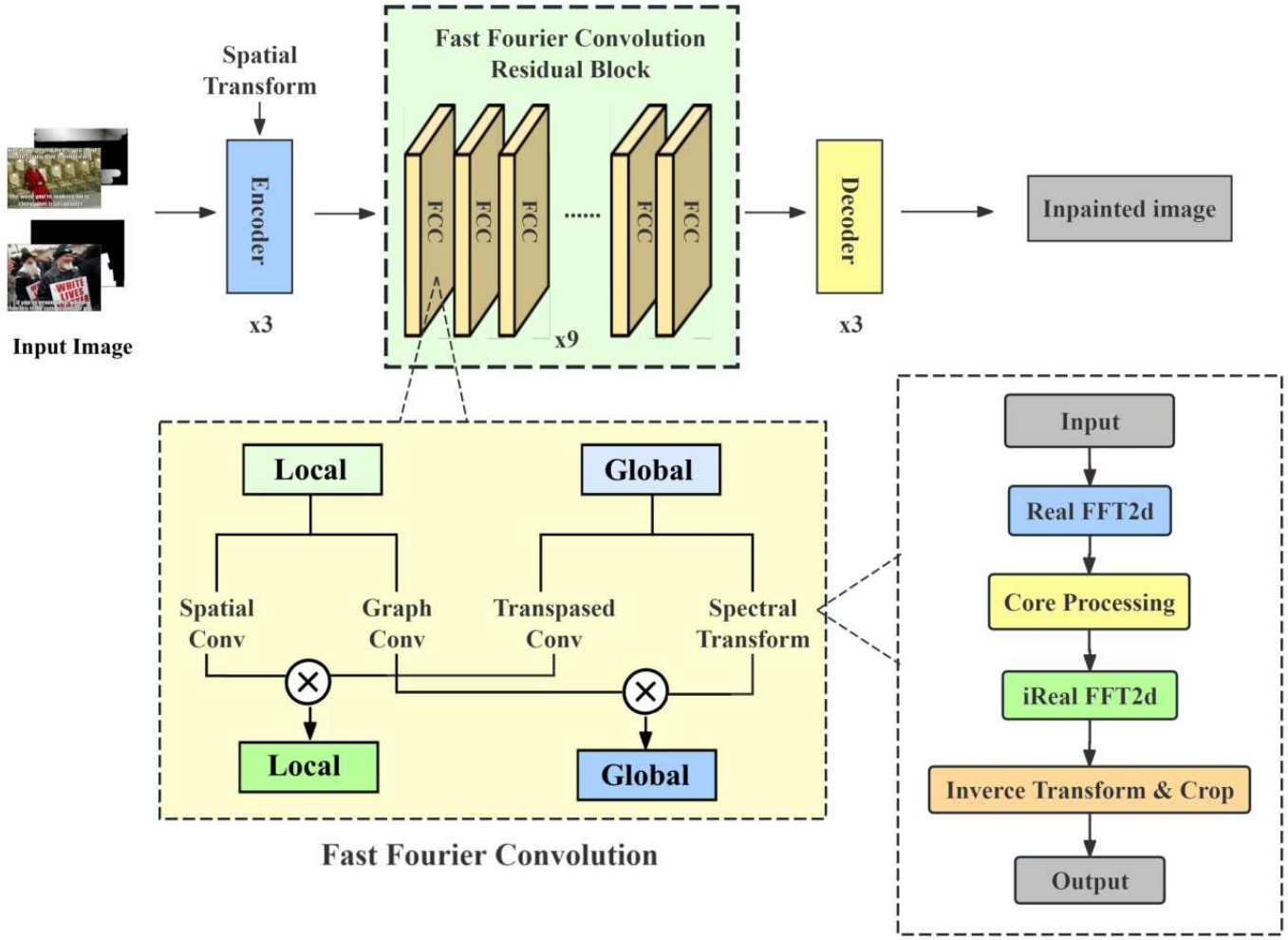
Generator:
1. Fast Fourier Convolution (FFC)
2. Local & Global modeling

GAN-based Discriminator： Enhance Visual Realism & Perceptual Quality

2.1 Big-LaMa Technology & Benefits

Local branch:
Patial convolution to recover fine-grained details

Global branch:
Spectral transformation to capture long-range dependencies.



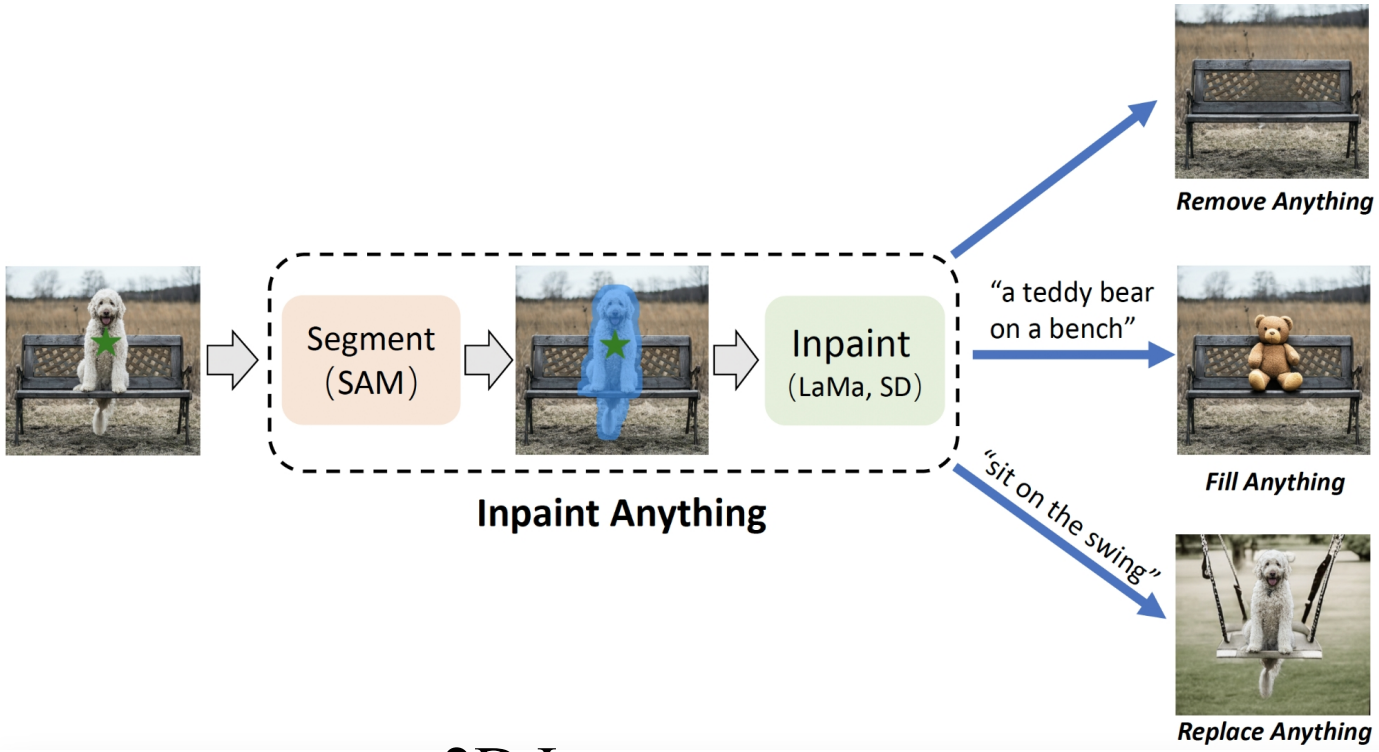
The structure of FCCGenerator

2.2 Big-LaMa Performance in 2D Image Inpaint

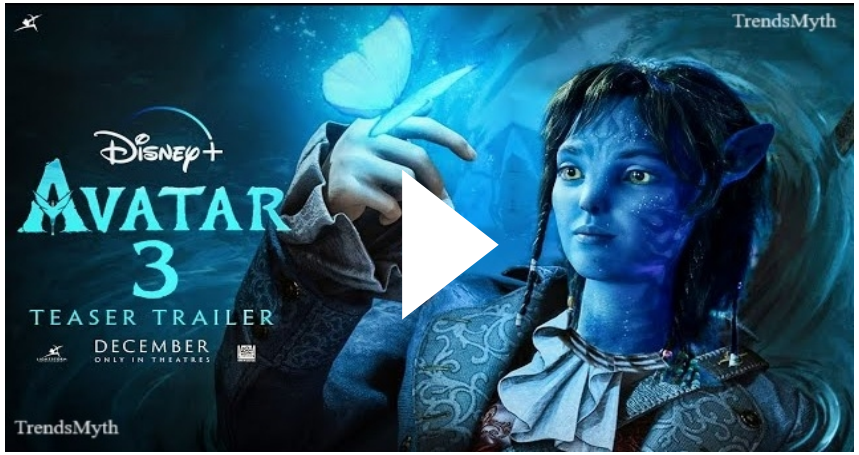


Is Big-LaMa Good Enough for Video Field?





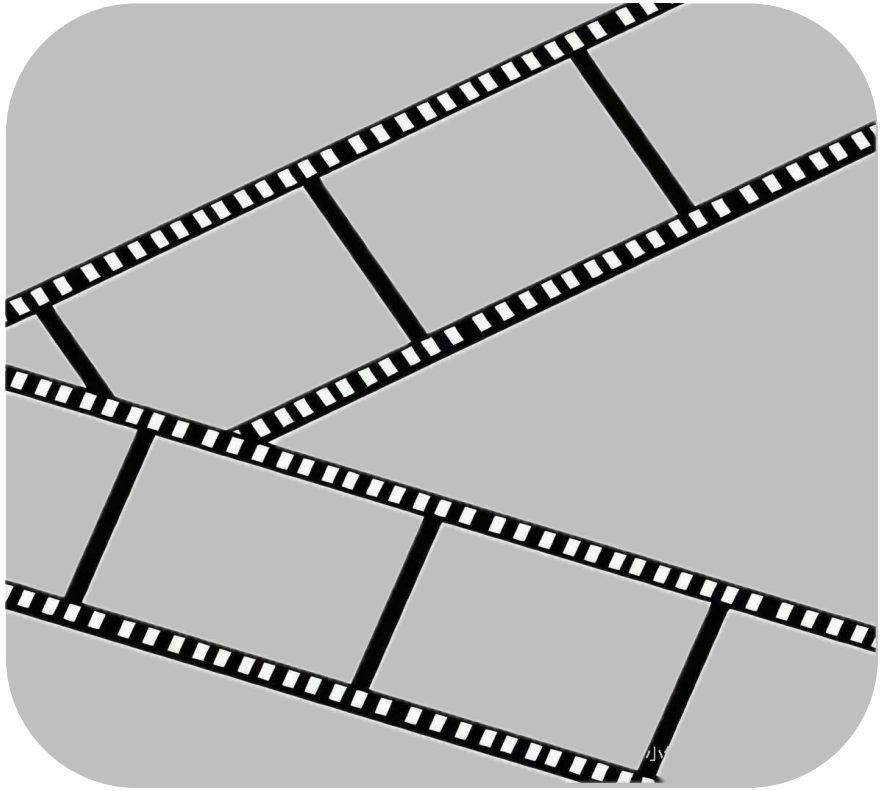
2D Image



Video

Idea: Apply Big-LaMa in each key frame & Combine

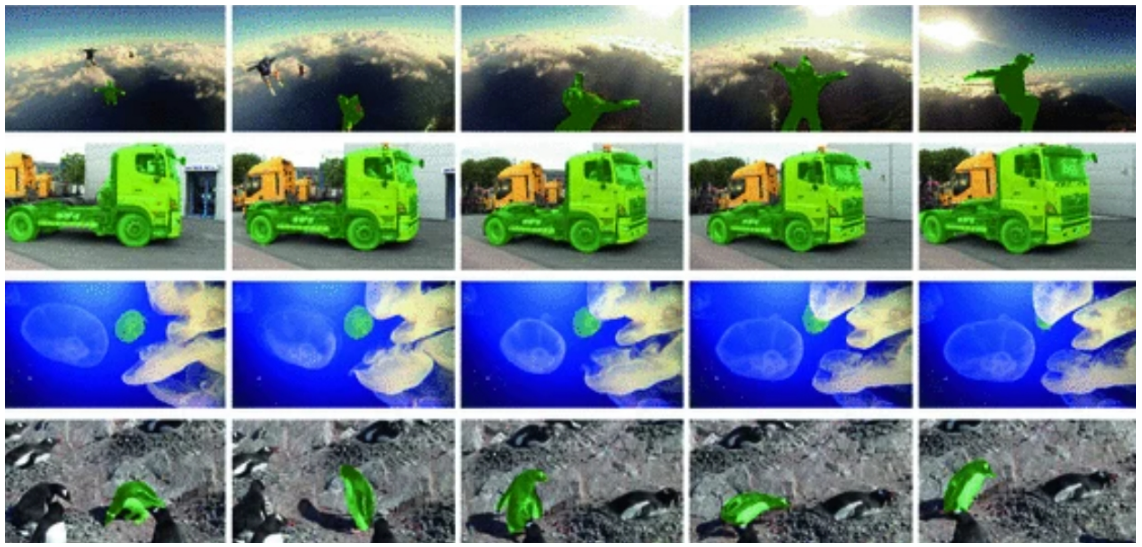
Image



First Step: New Dataset Fitting for Specific Scenes

Limitation of RVOS Dataset: **Short-term, Low-resolution & Simple Semantics**

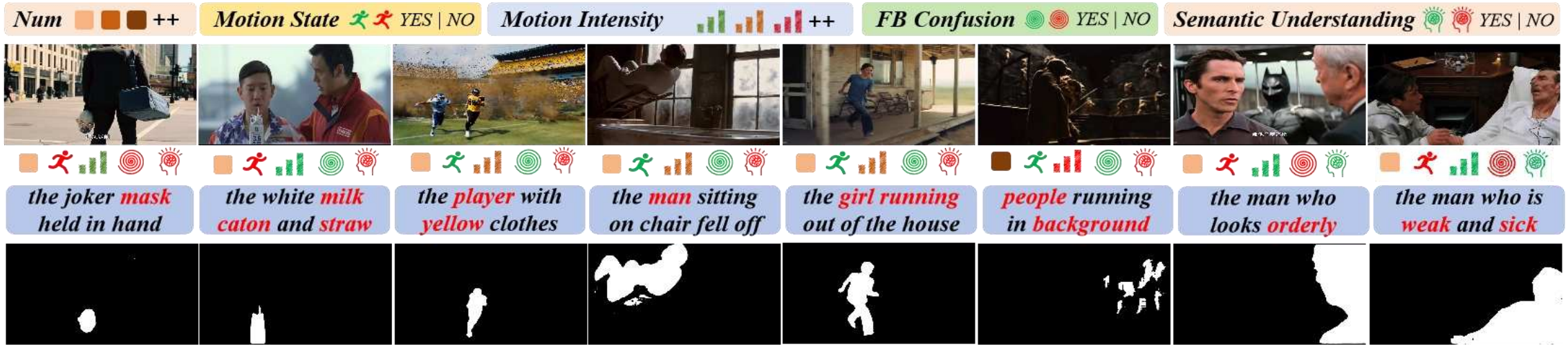




https://davischallenge.org/davis2016/code.html
https://link.springer.com/chapter/10.1007/978-3-030-01228-1_36

Require Dataset Fitting for Specific Scenes !!!

First Step: New Dataset Fitting for Specific Scenes



Number of Object (num)

Motion State

Motion Intensity

Foreground and Background Confusion

Semantic Understanding

Five Different Domains

Experiment 0: The Failure of Big-LaMa

Control Variables: To eliminate influence of mask, use Optimized P-MASk Module for generation.

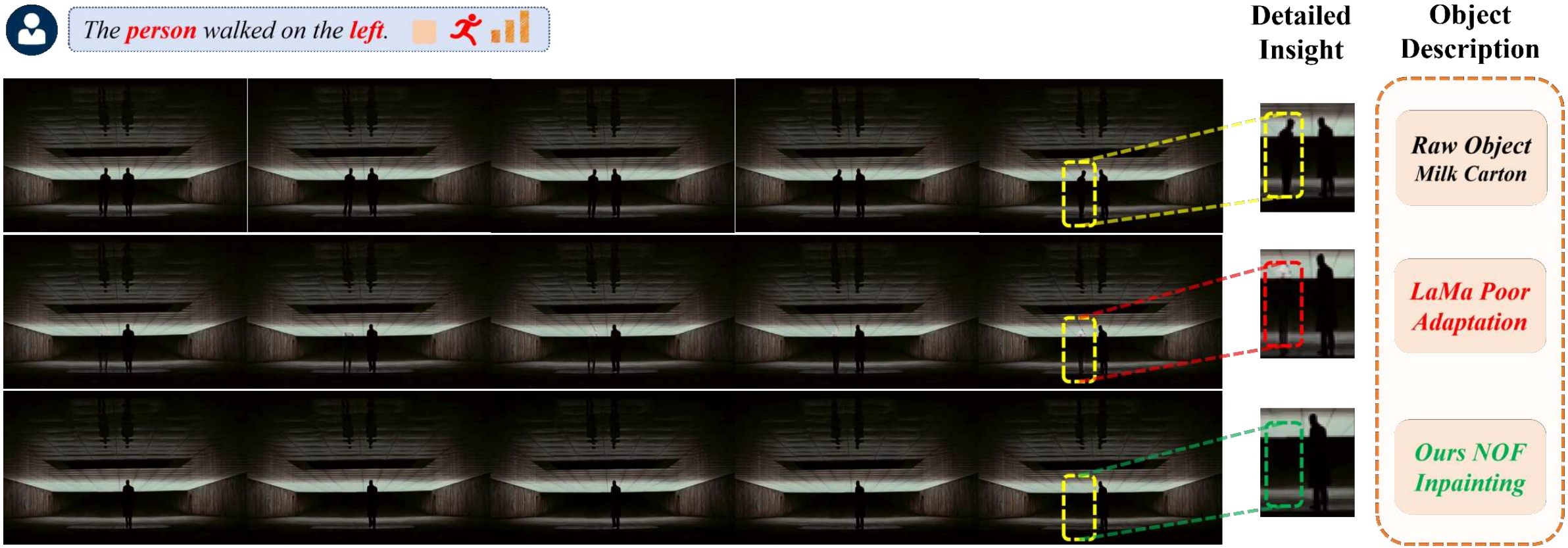


One obvious problem is **Time Issues !!!**

Time-consuming & Low-efficiency

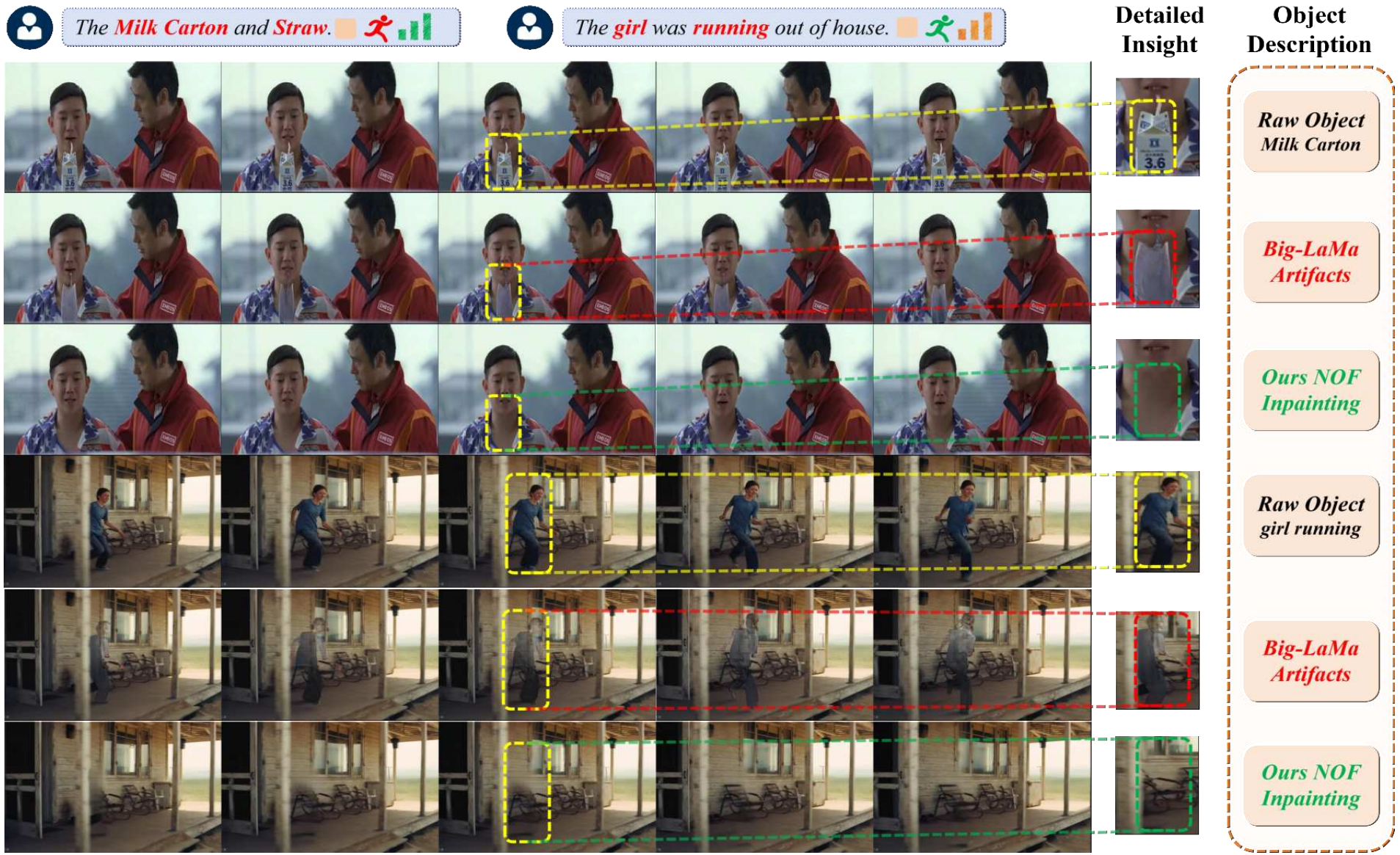
Experiment 0: The Failure of Big-LaMa

Failure1: Poor Adaptability



The model's repair performance is unsatisfactory in situations with high noise levels, weak lighting, drastic lighting changes, or highly complex backgrounds.

Failure 2： Artifact Problem



Fast-move targets

↓

Big-LaMa

↓

Capture temporal consistency ×

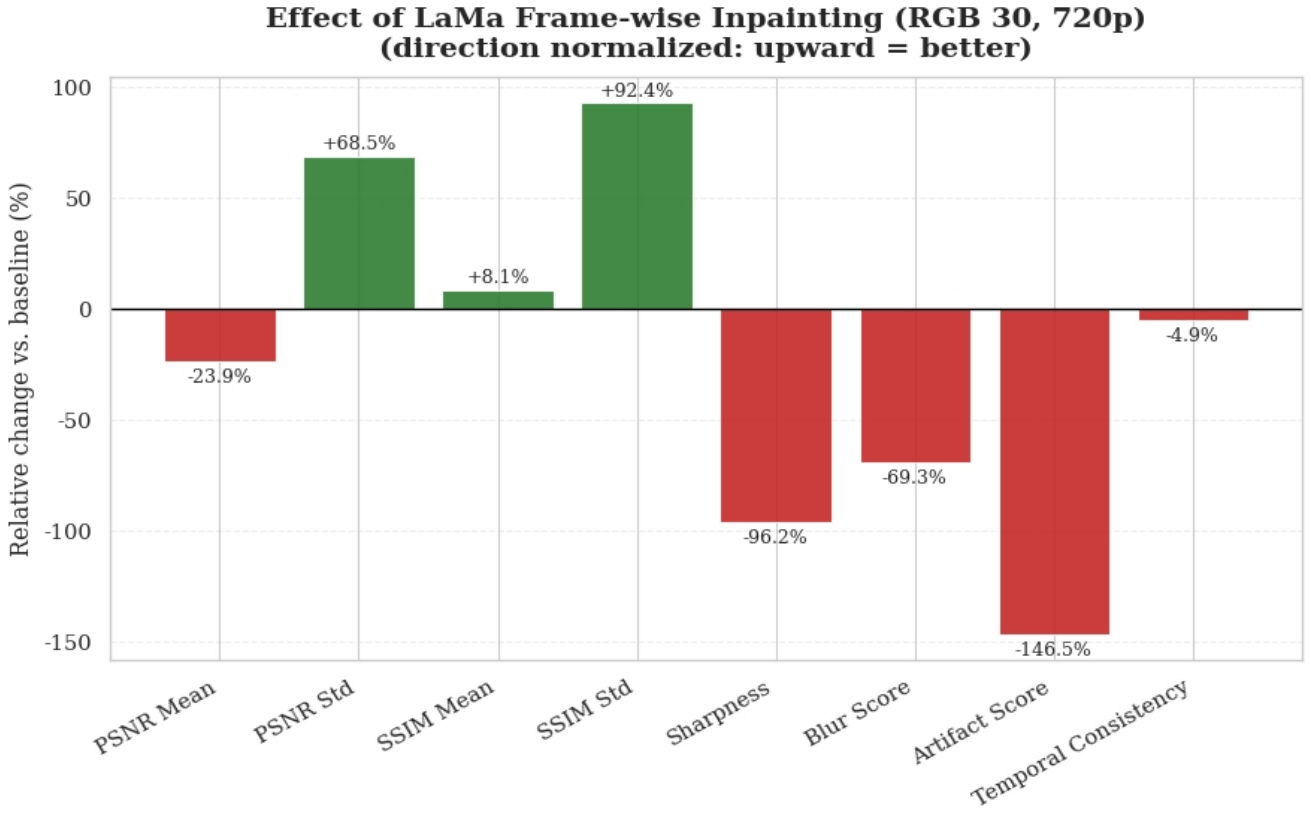
↓

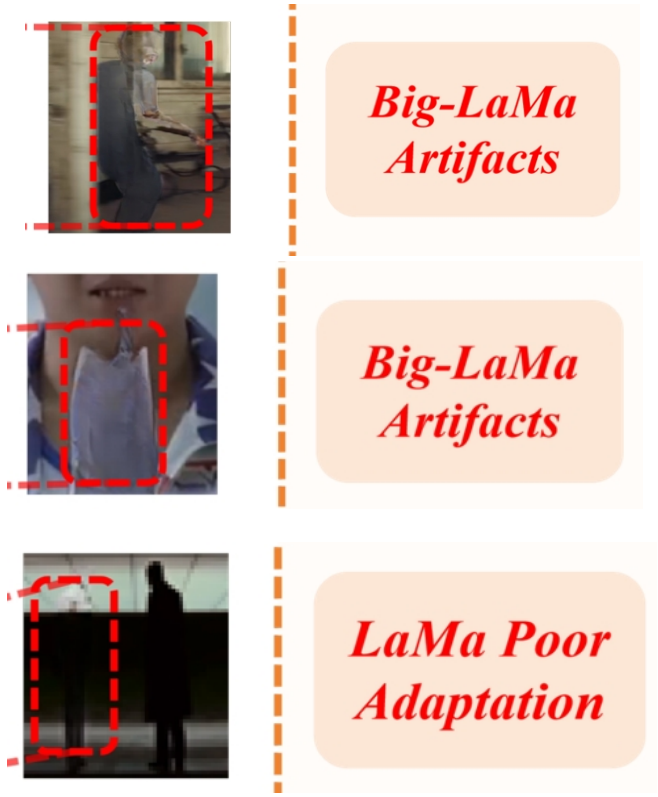
Unnatural transitions OR Artifact

Severe problems: Inter-frame Restoration Quality, Temporal Inconsistency & Artifact Issues

(Δ Artifact Score = −146.5%)

Modality & Fps	PSNR (Mean) / dB ↑	PSNR (Std) / dB ↑	SSIM (Mean) d/B ↑	SSIM (Std) d/B ↑	Sharpness Score ↑	Blur Score ↑	Artifact Score ↑	Temporal Consistency ↑	Runtime ↓ (s/frame)
RGB 30 (720p)	29.8239 (-9.3684)	2.473 (-5.3802)	0.956 (+0.0719)	0.0082 (-0.1001)	2.2208 (-55.7792)	48.8414 (+20.0002)	52.4213 (+31.1528)	78.8394 (-4.1007)	11.5646 (+11.2205)





PART 03

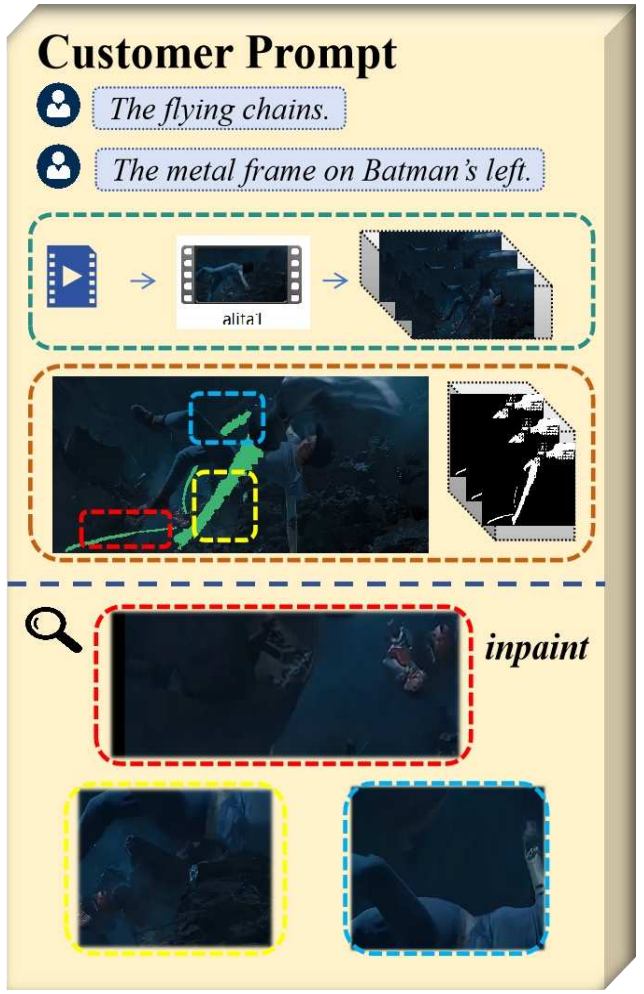
PEANUT in Video World

Presentor： Li Minghao (2330034027)

PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field

PEANUT : Three-Stage Cascade Pipeline





Prompt-based Mask Generation

P-MASk

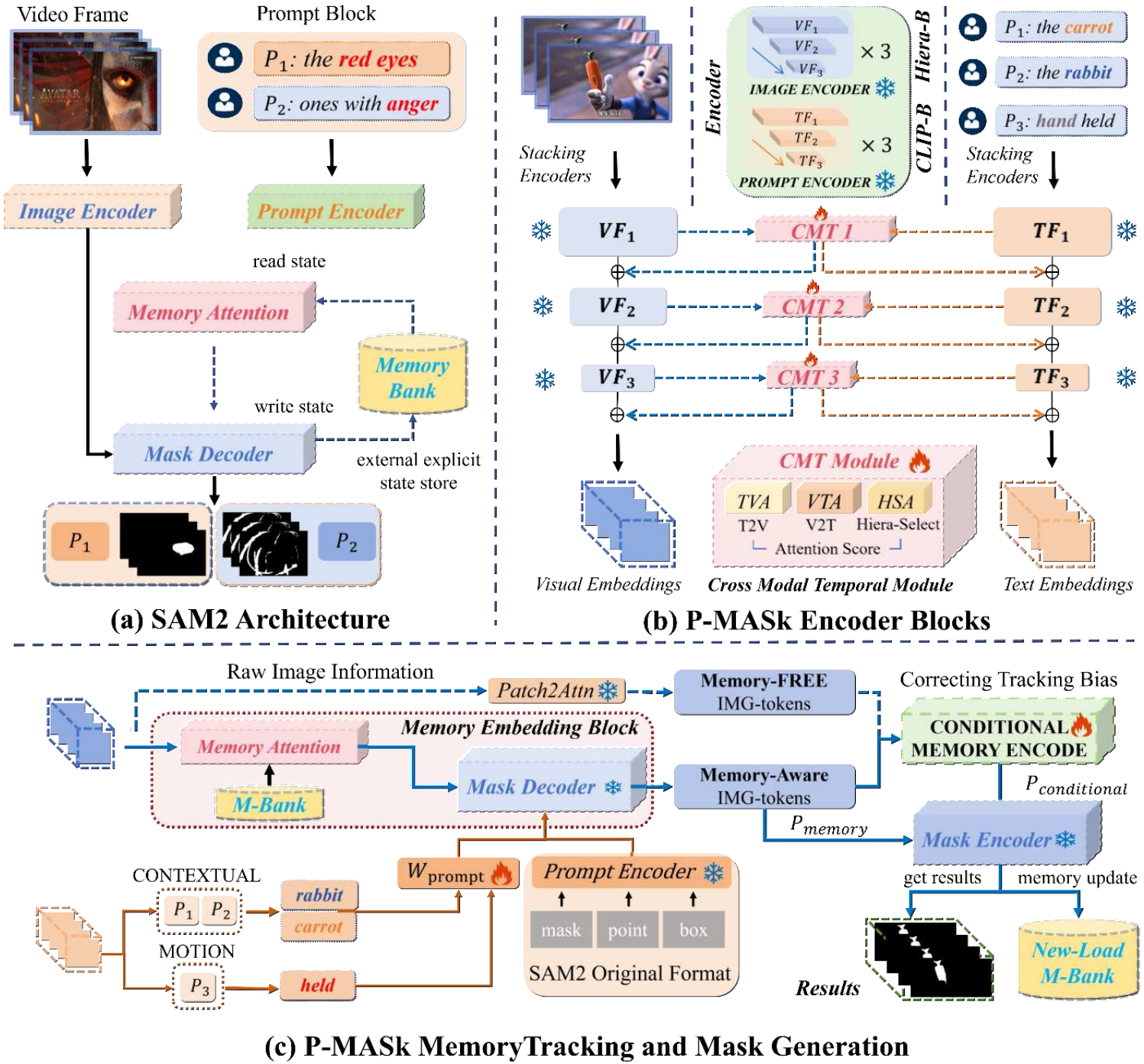
Inpaint & Enhance Block

NOF-Eraser — Inpaint

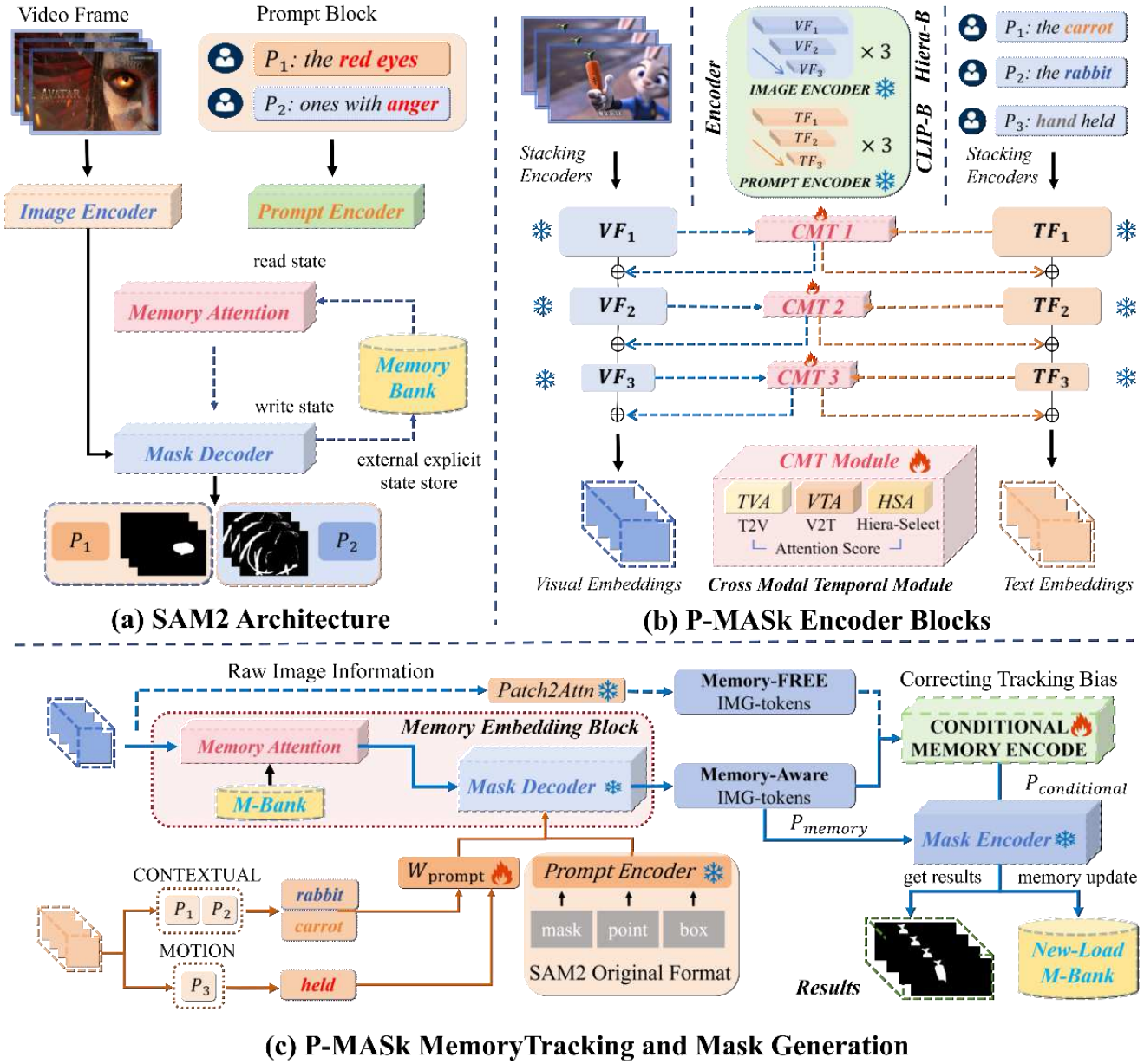
UR-Net — Enhance

Denise Net — Denoise

Module 1 P-MASk:
Prompt-Guided Mask Generation



Module1: P-MASk (Prompt-Guided Mask Generation)





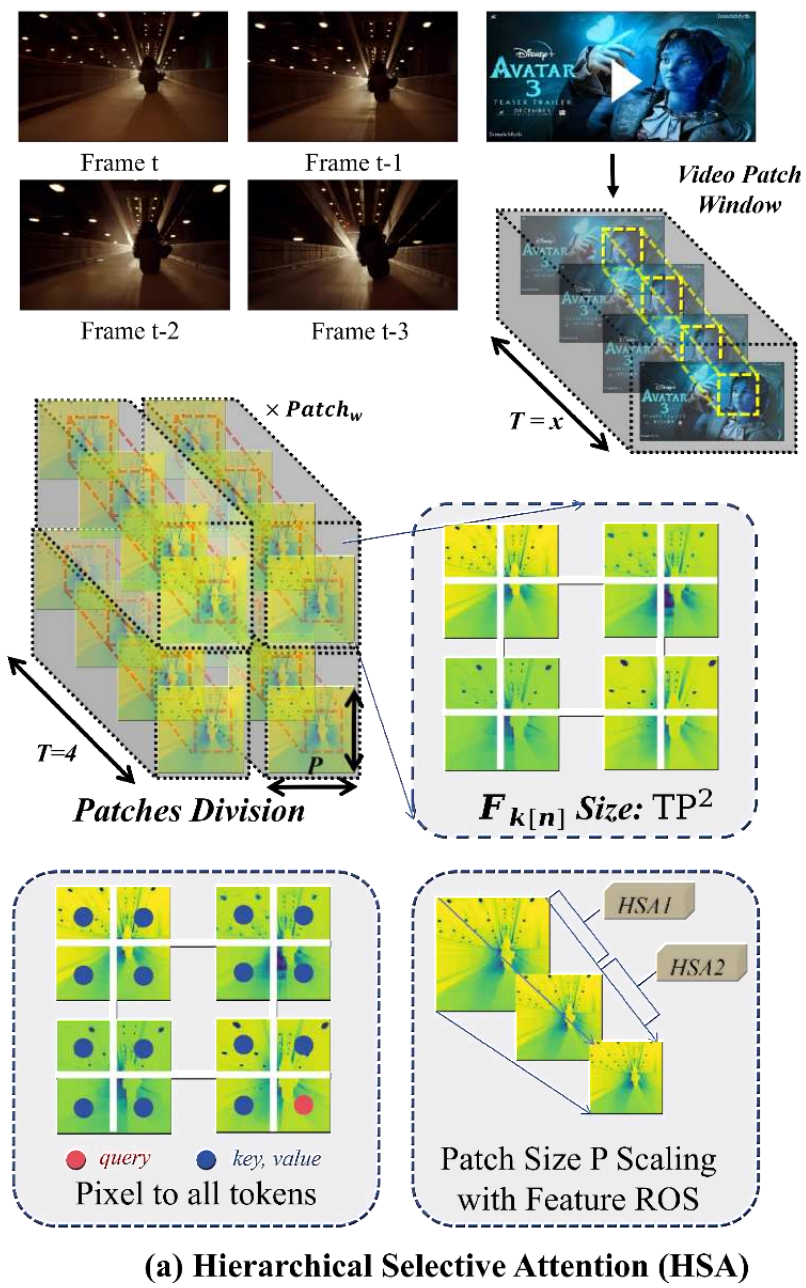


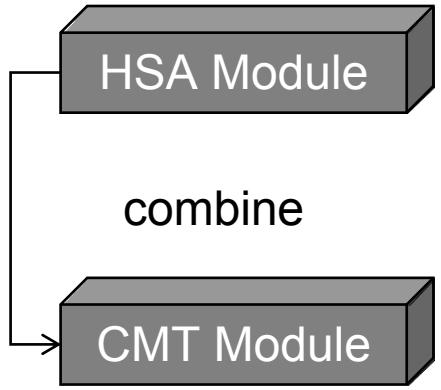
Input Output

Key Components:

(1) Encoder Block (stakcing)
- CMT (Cross-modal timing adapter)
- IMG & Video Fusion
- HSA

(2) Memory Tracking Bias Block
- CME (Eliminate bias in long-term tracking)





Follow the SAM2 Official Implementation, we follow the base for lightweight deployment:

1) Visual Encoder: Hiere-B
2) Text Encoder: CLIP-B

Key idea:

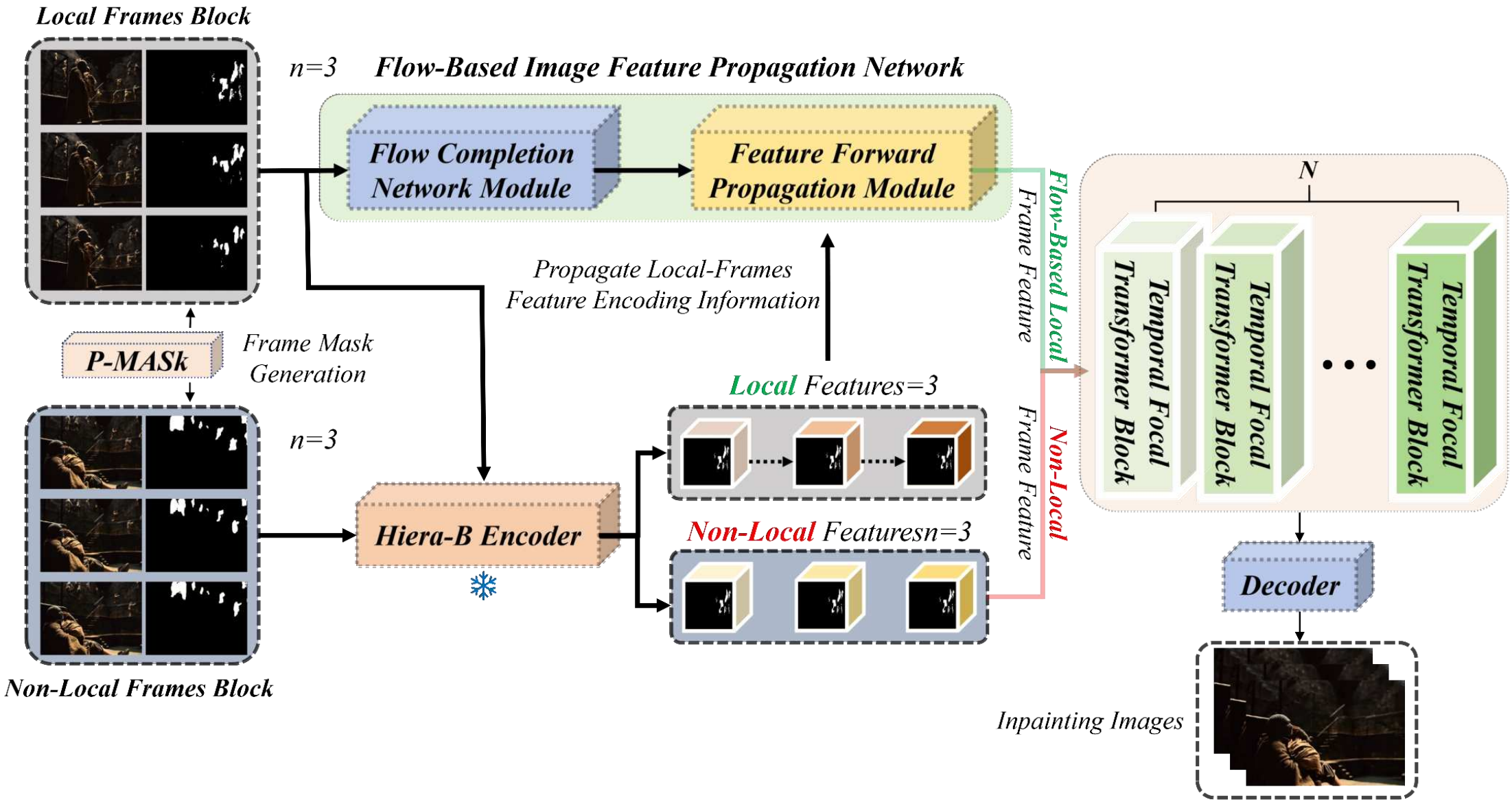
Basic Cross-modal interaction

+

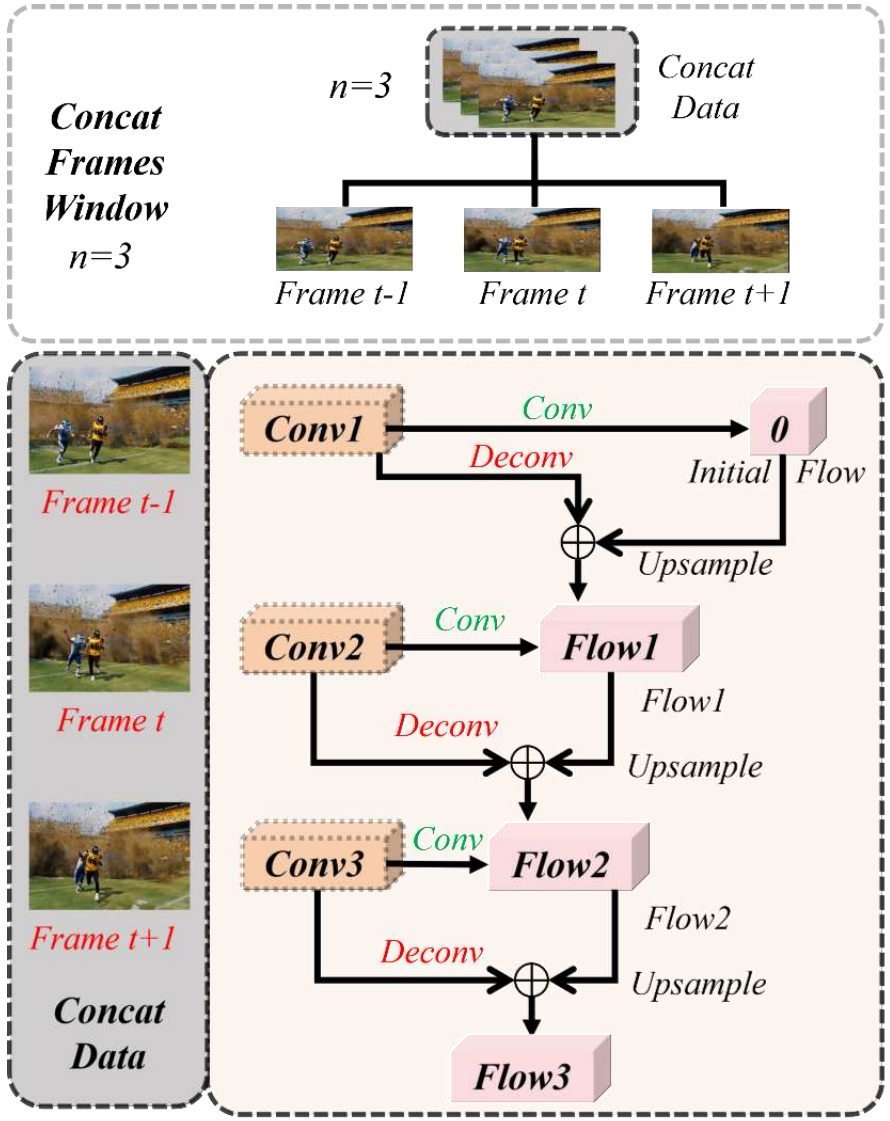
HSA Mechanism

(Progressively focusing on regions of different scales, the model is helped to effectively handle different details in long time series)

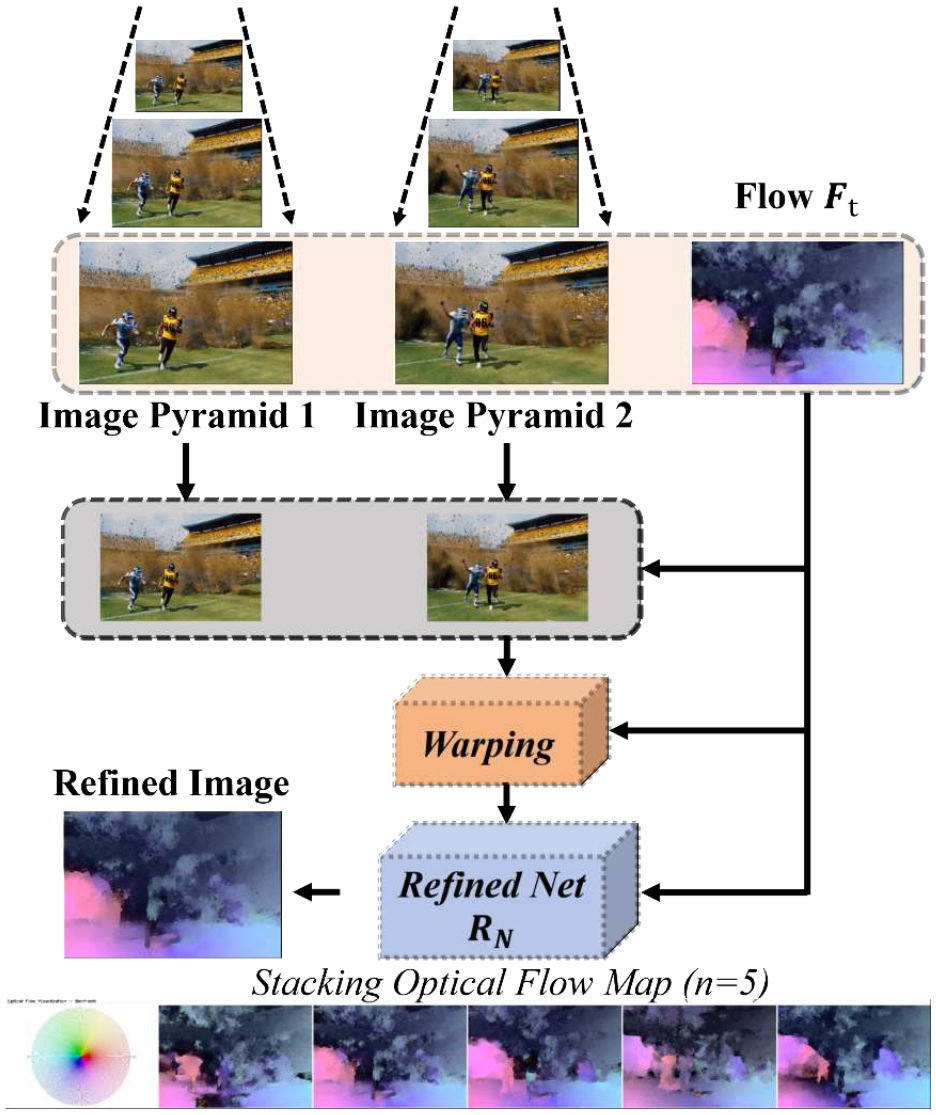
Module 2 NOF-Eraser (Neural Optical Flow-Based Inpaint)



Flow-Based Backbone SpyNet: Classic Optical Flow Estimation Model



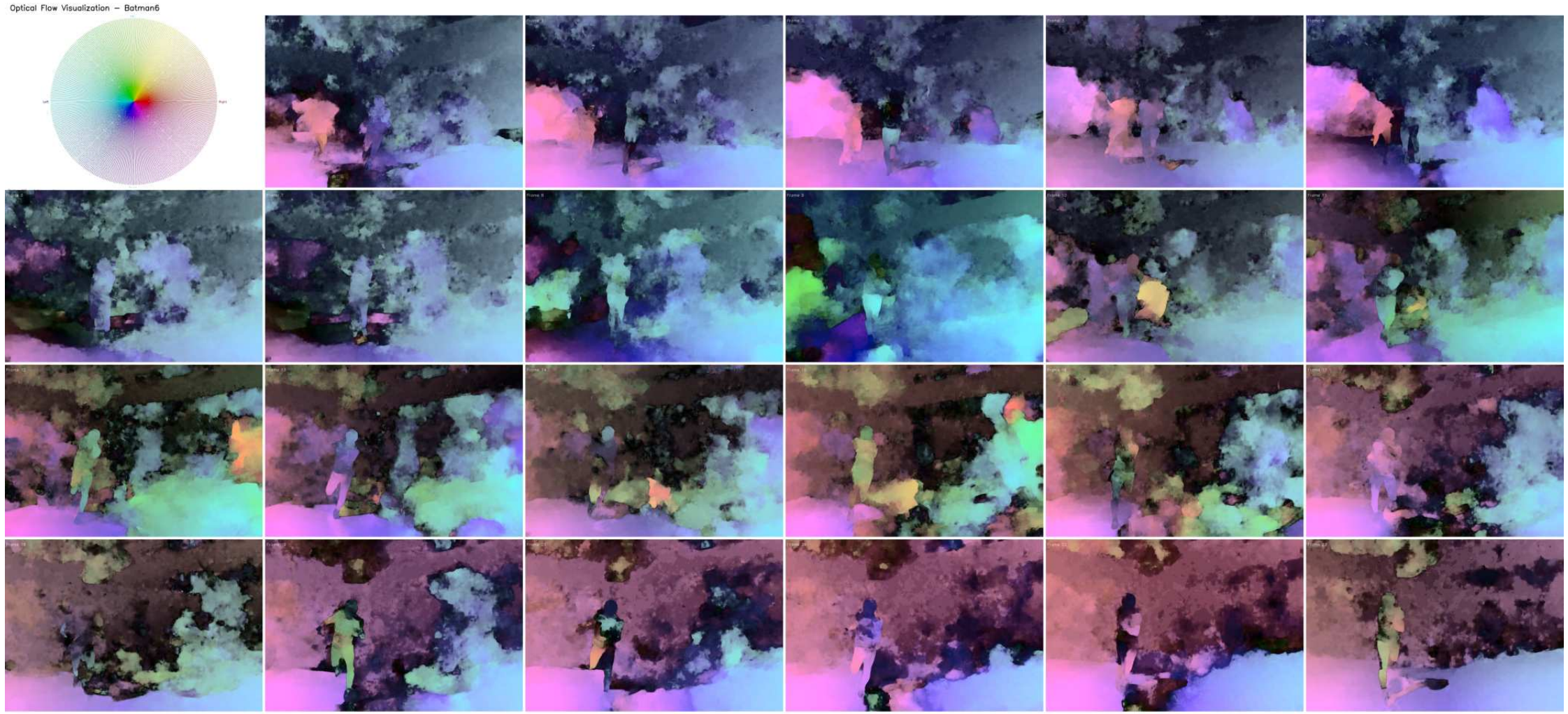
(a) Spatial Pyramid Block of Flow Estimation



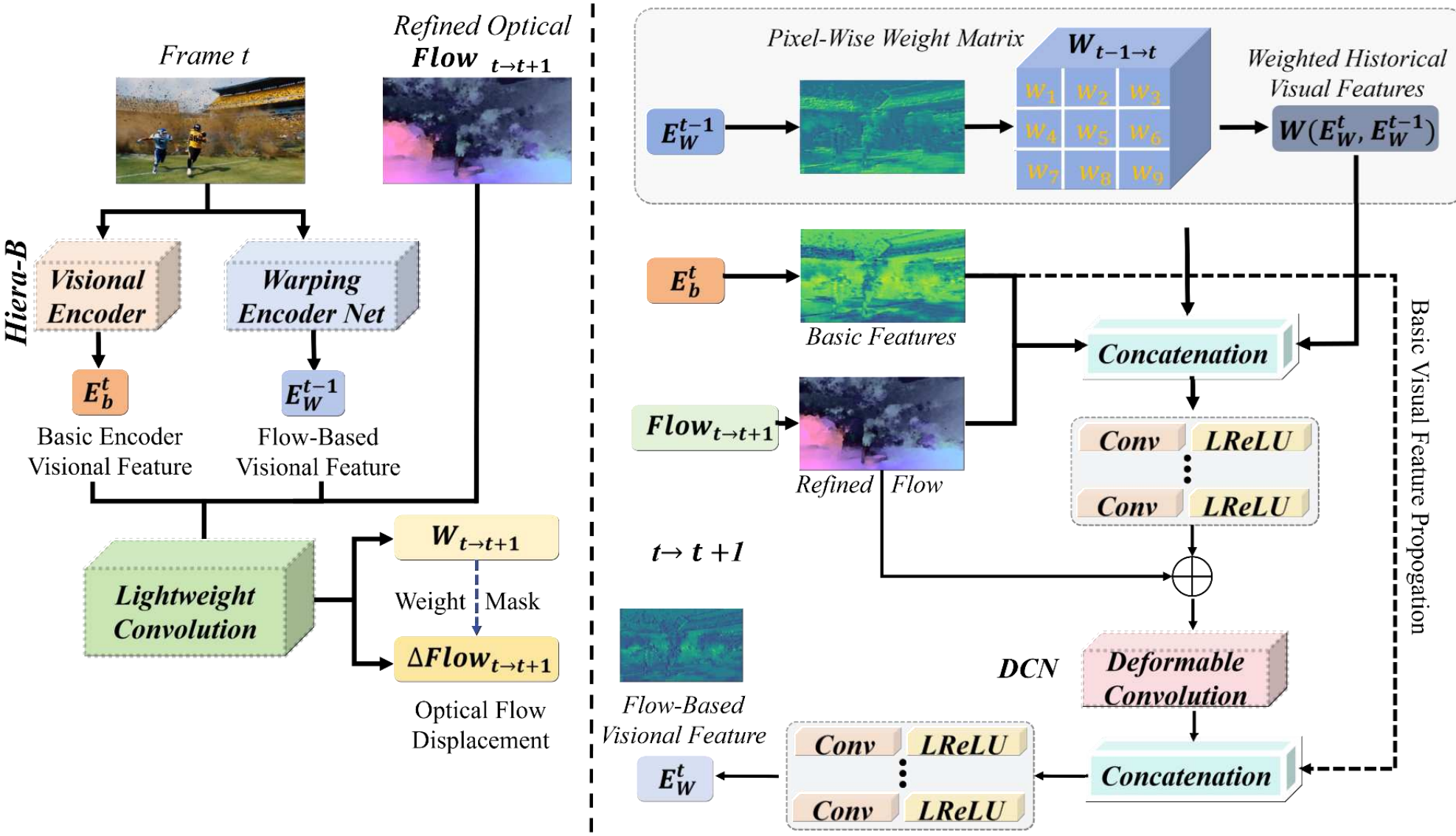
(b) Spatial Pyramid Network and Refined Block

An example of our self-created dataset:

Batman6.mp4 Flow Estimation Process



Visualzation of Optical Flow Estimation in Batman6.mp4



Feature Forward Propagation Module:

Two Steps:

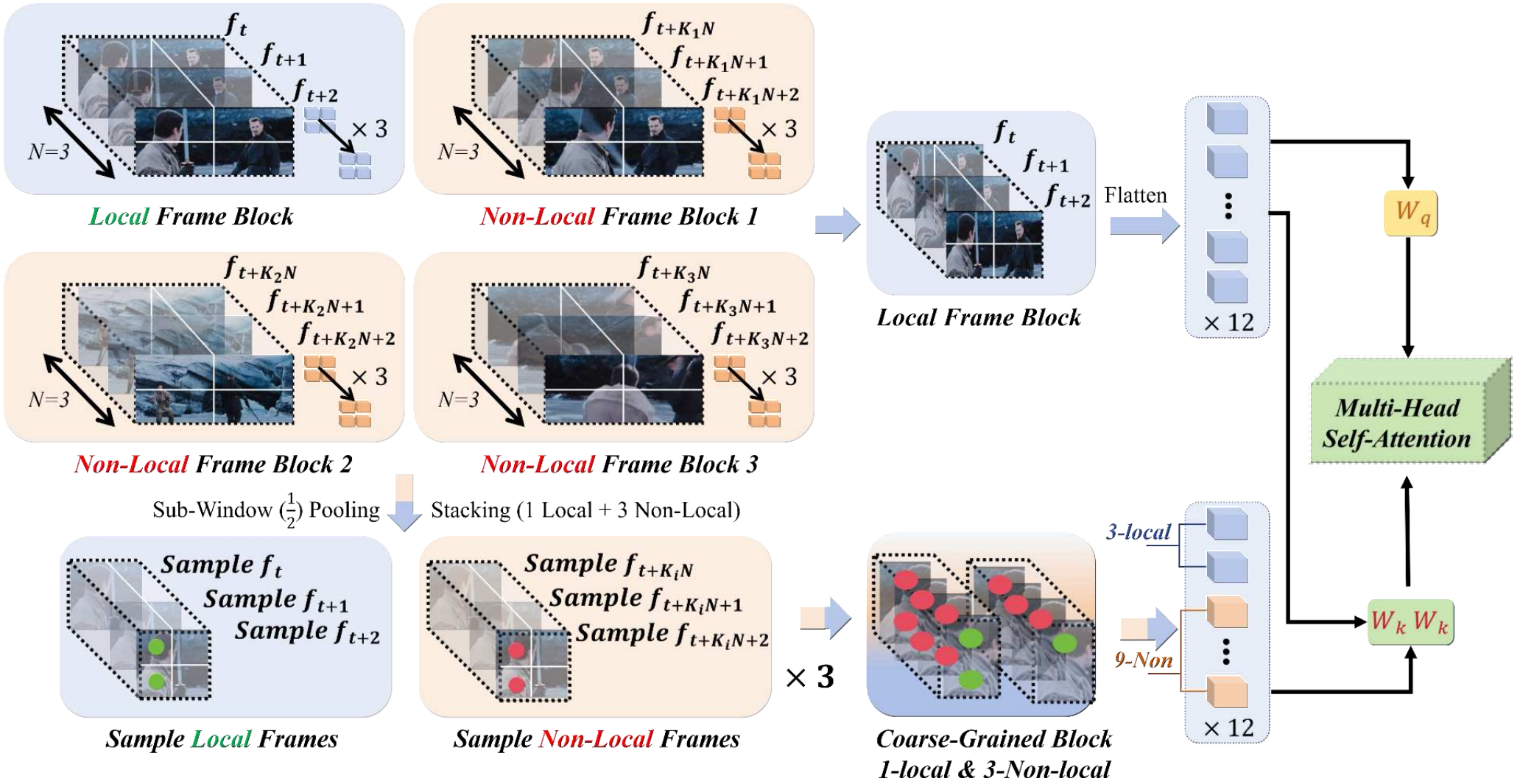
1.CNN prepared for DCN Input

2.Feature Encoding and combined with Estimated Flow

(a) Lightweight Convolution Network Prepare the Prediction for DCN Input

(b) Feature Forward Propagation Optical-Flow Based Module: the opposite direction for Backward Flow Propagation

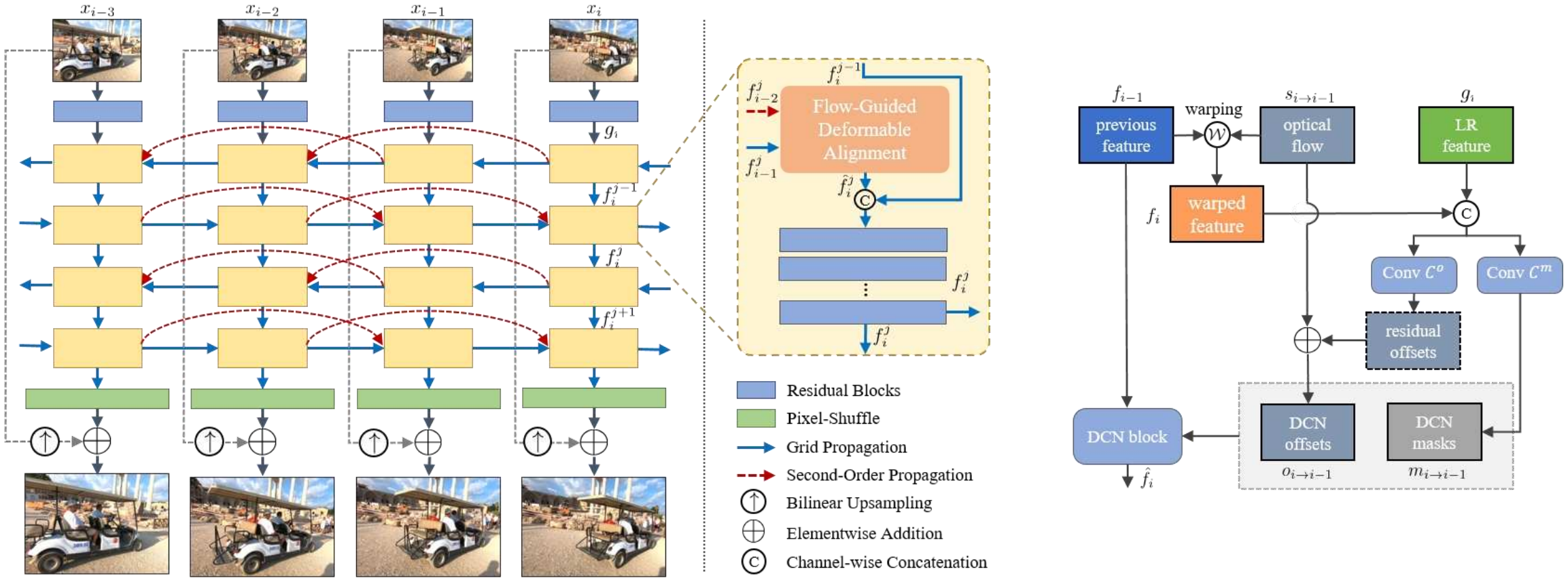
Temporal Focal Transformer Block



Gather the information about both local-frame and non-local frame by s、

Simple multi-head self-attention.

Module 3 UR-Net: Ultra-Resolution & Coherence



We directly load in this block from the official implementation allows us to directly import the library for video super-resolution optimization.

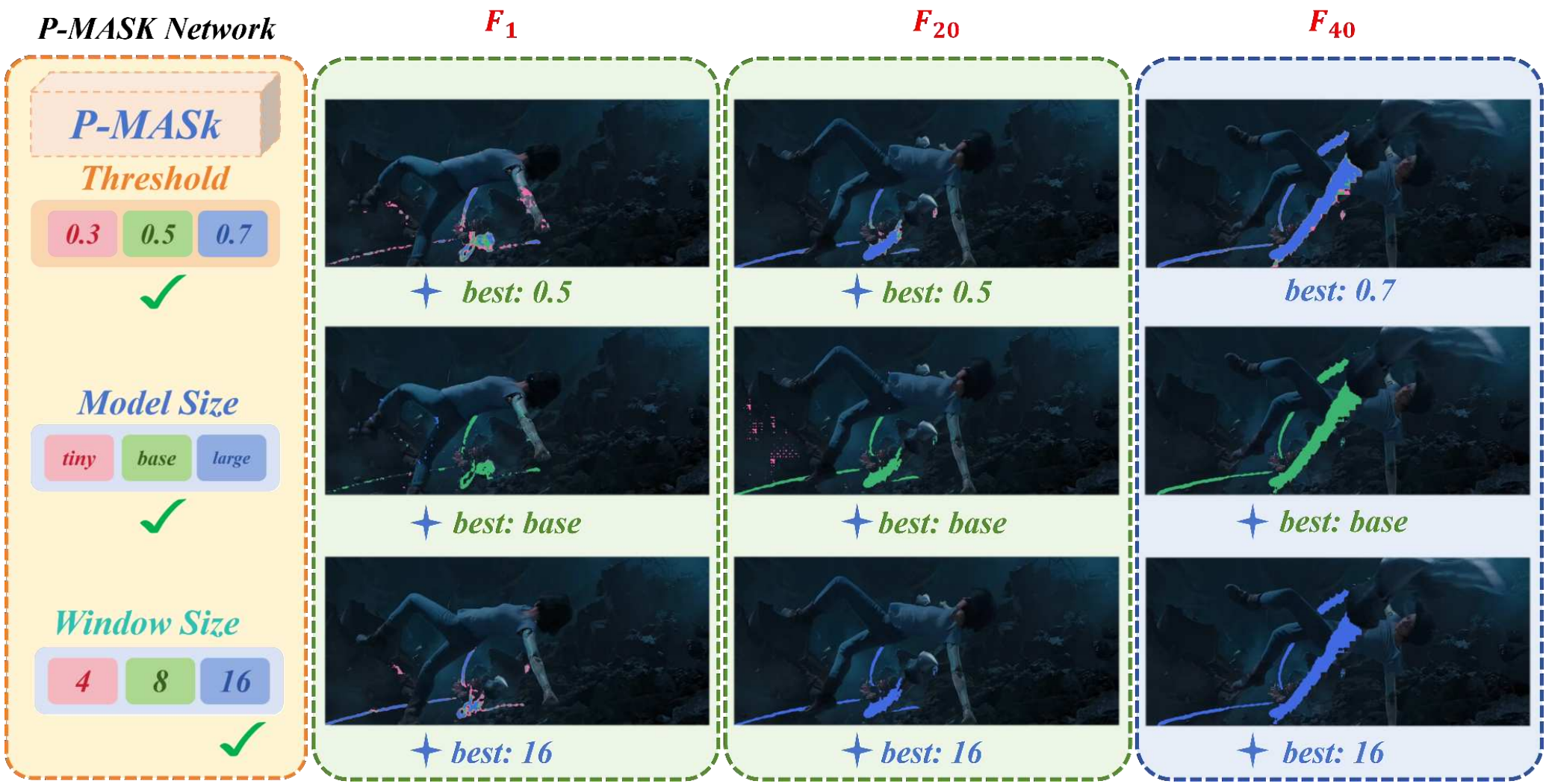
PART 04

Experiment & Evaluation

Presentor： Li Minghao (2330034027)

PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field

Experiment 1: P-MASk Hyperparameters Selection



Visualization of Different Hyperparameters Selction

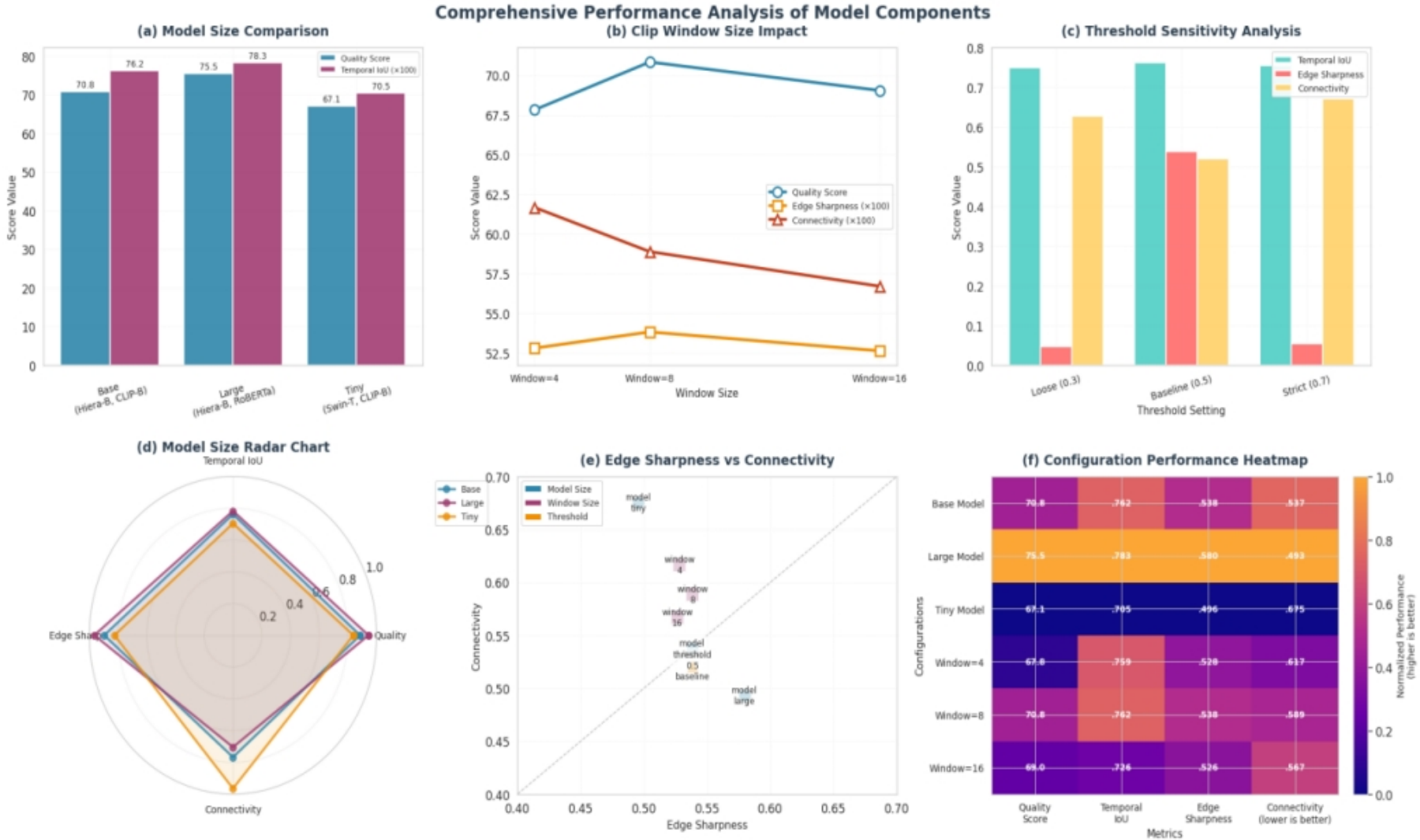
Experiment 1: P-MASk Hyperparameters Selection

				Model Component				
Hyperparameters	**Test Name**	**Best Balance**	**Backbone**	**Visual Encoder**	**Text Encoder**	**ClipWindow**	**Threshold**	**Modality & Fps**
Model Size	model_base	√	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	8	baseline (0.5)	RGB 30 (720p)
	model_large	×	SAM 2 (2024)	Hiera-B (89M)	RoBERTa	8	baseline (0.5)	RGB 30 (720p)
	model_tiny	×	SAM 2 (2024)	Swin-T (28M)	CLIP-B	8	baseline (0.5)	RGB 30 (720p)
Window Size	window_16	×	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	16	baseline (0.5)	RGB 30 (720p)
	window_4	×	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	4	baseline (0.5)	RGB 30 (720p)
	window_8	√	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	8	baseline (0.5)	RGB 30 (720p)
Threshold	threshold_0.3_loose	×	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	8	loose (0.3)	RGB 30 (720p)
	threshold_0.5_baseline	√	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	8	baseline (0.5)	RGB 30 (720p)
	threshold_0.7_strict	×	SAM 2 (2024)	Hiera-B (89M)	CLIP-B	8	strict (0.7)	RGB 30 (720p)

Quality Score	**Geometric Structure Metrics**		
PSNR↑ : SSIM↑ : Sobel_sharpness↑ : TC↑ = 6:4:3:7	**Temporal IoU ↑**	**Edge Sharpness ↑**	**Connectivity ↑**
70.8393	0.7623	0.5383	0.5213
75.4921	0.7829	0.5795	0.4931
67.0836	0.7046	0.4957	0.6753
69.0481	0.7257	0.5264	0.5671
67.8301	0.7595	0.5281	0.6167
70.8393	0.7623	0.5383	0.5213
60.9371	0.7492	0.047	0.6284
70.8393	0.7623	0.5383	0.5213
68.4839	0.7543	0.054	0.6714

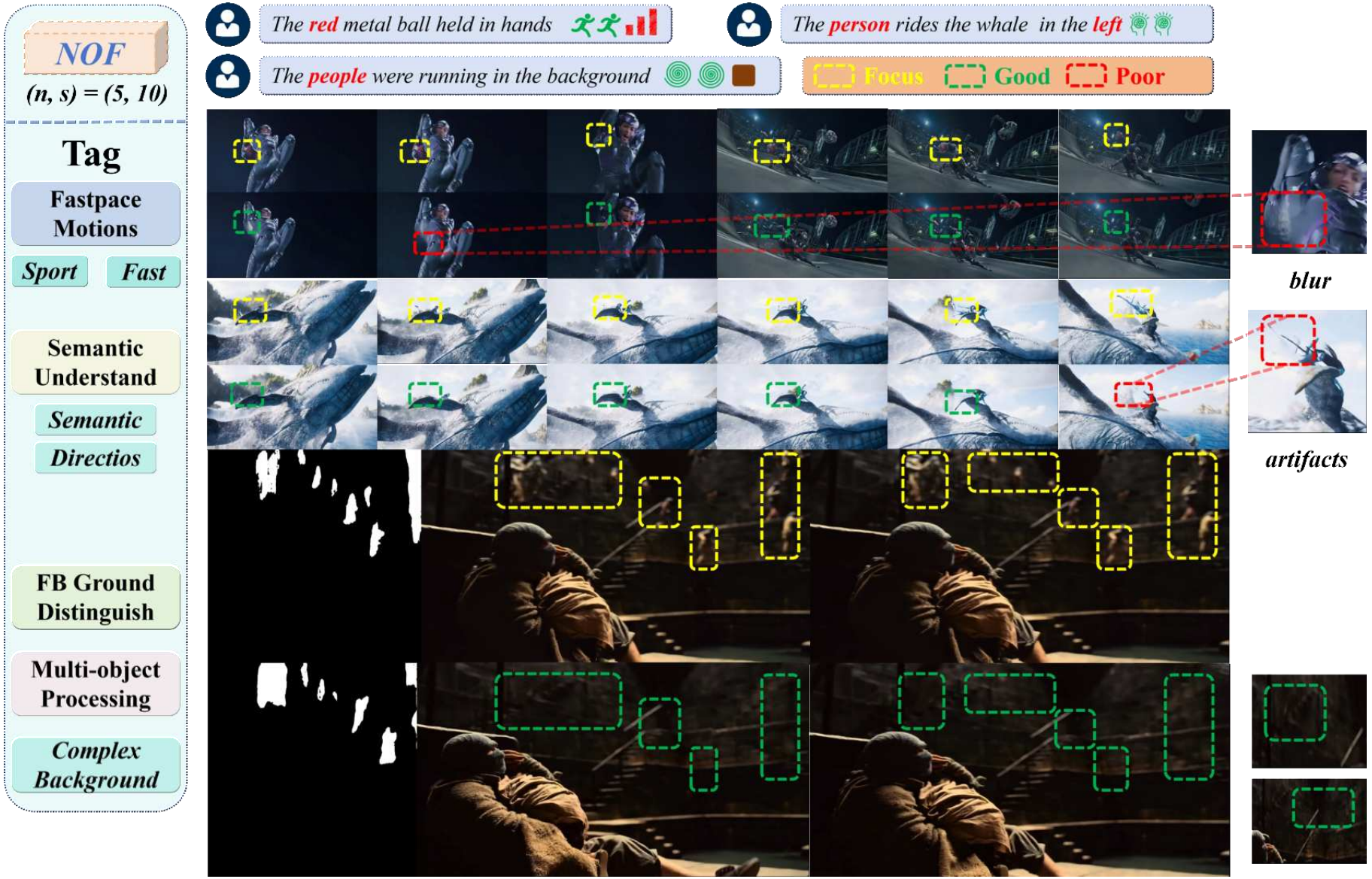
Statistic Information of Hyperparameters Selection Problems

Experiment 1: P-MASk Hyperparameters Selection



Visualization of Hyperparameters Selection Problems

Experiment 2: NOF-Eraser Hyperparameter Selection

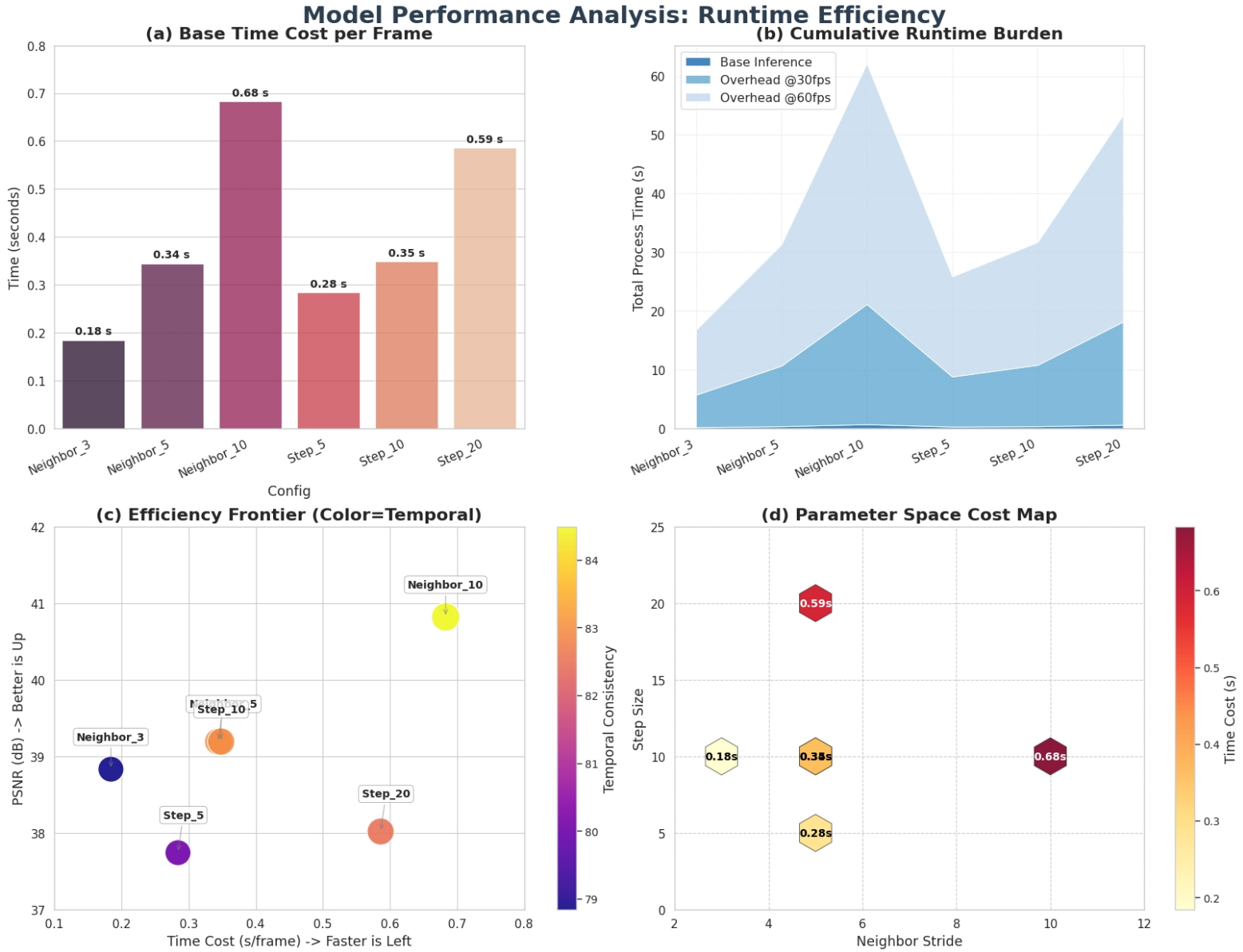


Experiment 2: NOF-Eraser Hyperparameter Selection

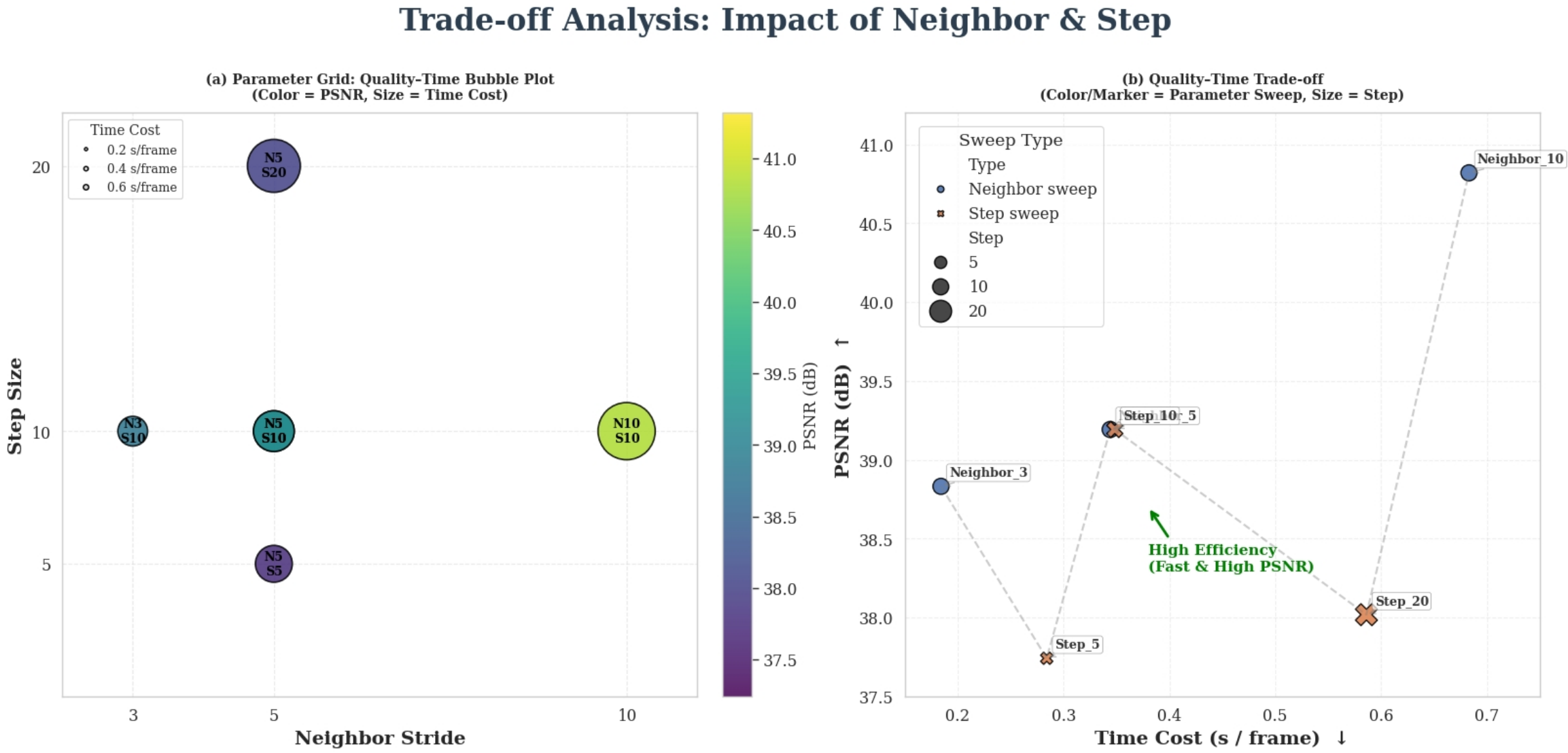
Model Settings			Best Balance	Video Settings	
Configuration	**Neighbor Stride**	**Step**	**choice**	**Modality & Fps**	**Reference Number**
neighbor_3	3	10	×	RGB 30 (720p)	-1 (default all)
neighbor_5	5	10	√	RGB 30 (720p)	-1 (default all)
neighbor_10	10	10	×	RGB 30 (720p)	-1 (default all)
step_5	5	5	×	RGB 30 (720p)	-1 (default all)
step_10	5	10	√	RGB 30 (720p)	-1 (default all)
step_20	5	20	×	RGB 30 (720p)	-1 (default all)

Image Fidelity Indicators				Perceived Quality and Geometric Indicators			Temporal & Geometric Metrics	Runtime ↓ (s/frame)	Runtime ↓ (s / video second)		
PSNR (Mean) / dB ↑	**PSNR (Std) / dB ↑**	**SSIM (Mean) d/B ↑**	**SSIM (Std) d/B ↑**	**Sharpness Score ↑**	**Blur Score ↑**	**Artifact Score ↑**	**Temporal Consistency ↑**	**Time Cost ↓**	**30 fps**	**60 fps**	**120 fps**
38.8319	9.4814	0.8718	0.1083	58	30.4814	22.4213	78.8394	0.1842	5.526	11.052	22.104
39.1923	7.8532	0.8841	0.0984	64	28.8412	21.2685	82.9401	0.3441	10.323	20.646	41.292
40.8194	10.7418	0.8765	0.0921	69	29.4673	22.8913	84.4885	0.6831	20.493	40.986	81.972
37.7419	8.7414	0.8673	0.0905	58	29.9841	21.4254	80.0348	0.2841	8.523	17.046	34.092
39.1923	7.8532	0.8841	0.0984	64	28.8412	21.2685	82.9401	0.3441	10.323	20.646	41.292
38.0184	9.8414	0.8692	0.1056	63	28.8955	21.2676	82.4819	0.5862	17.586	35.172	70.344

Experiment 2: NOF-Eraser Hyperparameter Selection



Experiment 2: NOF-Eraser Hyperparameter Selection



PART 05

Conclusion

Presentor： Tu Yixin (2330034048)

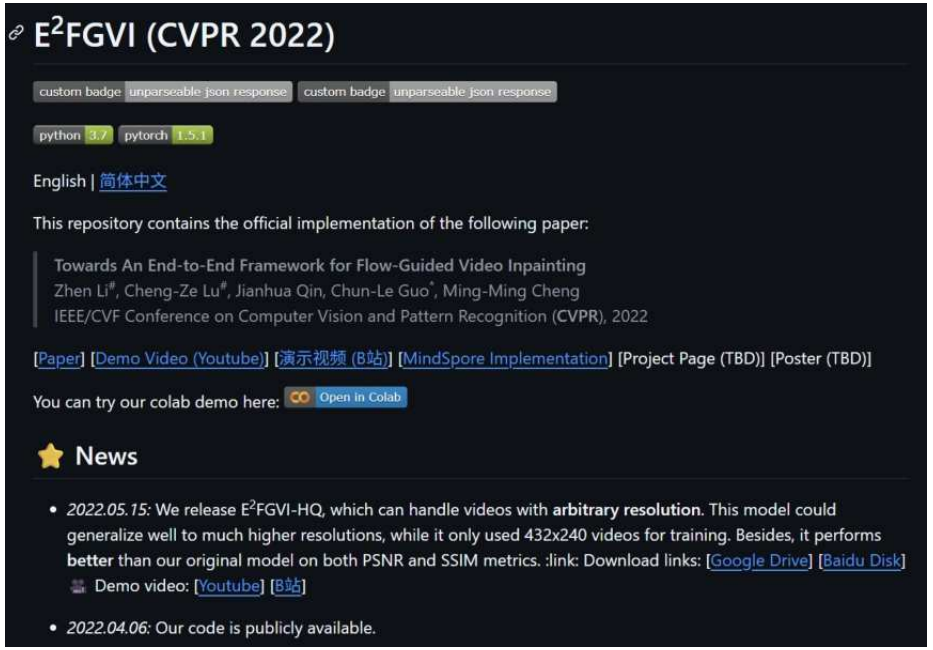
PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field

5.1 Innovative Fusion

1. Prompt-Guided Optimization of an End-to-End Flow-Guided Video Inpainting Network

Limitations of Optical Flow Methods：

①**Manual generation of target masks** → Complex Operation & Low Accuracy
②**High precision required for each sampled frame** → Difficult to handle fast-paced scenes







5.2 Our Innovative Fusion

2. Track the right object !

We develop a module named Conditional Memory Encoder (CME).
In each frame, we produce 2 informed mask frames.



Original Sample Frame



Memory-Free Mask



Memory-Aware Mask

CME

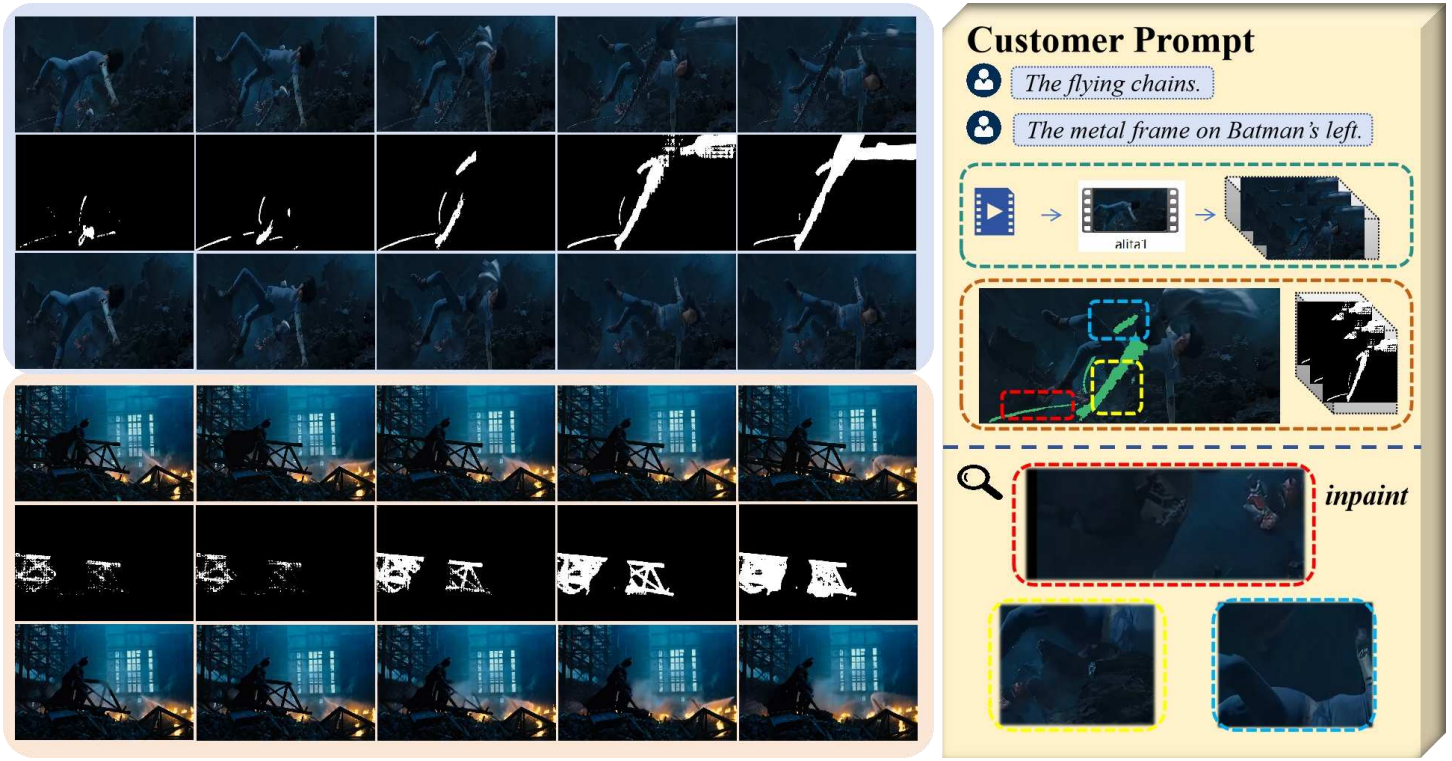
3.3 Our Innovative Fusion

3. Lightweight Model Deployment & Easy Implementation with Personal Data

We balance the Speed / Memory Hard-ware Demand / Quality

4. One-Stop Service (Direct Prompt2Results)

- Mask self-generation
- BasicVSR++ Algorithm to make super resolution
- Flow-based model with DL to enhance inpaint performance
- Denoise Module based on NLM algorithm



Our One-Stop Process

Thank you for your listening!

PEANUT: Prompt-Enhanced Ablation with Optical Flow-Based Neural Unit for Spatio-Temporal Consistency & VSR++ Clarity from IMG2Video Field 💫

Python v3.8 | PyTorch v2.6.0+cu124 | CUDA v12.4

TorchAudio v2.6.0+cu124 | TorchVision v0.21.0+cu124 | imageio-ffmpeg v0.6.0